Michigan Tech

Elliptical Orbits

Derivation from the relative displacement equation

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Outline

1. Use the central force to establish planar motion.
2. Write the relative displacement equation in polar coordinates.
3. Solve for the reciprocal separation as a function of angle.
4. Derive the ellipse and identify the energy condition.

Relative displacement equation

- Recall the relative displacement and total mass from the two-body reduction.

$$\boldsymbol{r} = \boldsymbol{r}_2 - \boldsymbol{r}_1 \qquad r = \|\boldsymbol{r}\| > 0 \qquad M = m_1 + m_2$$

- The isolated system satisfies

$$\boxed{\boldsymbol{r}'' = -GM\frac{\boldsymbol{r}}{r^3}}$$

- Write $k = GM > 0$. Then $\boldsymbol{r}'' = -k\boldsymbol{r}/r^3$.
- Keep $\mu = m_1 m_2/M$ for reduced mass; k is the gravitational parameter [1].

Angular momentum conservation

1. Define **Specific Angular Momentum**

$$\boldsymbol{h} = \boldsymbol{r} \times \boldsymbol{r}'$$

2. Differentiate and use the central acceleration.

$$\begin{aligned} \boldsymbol{h}' &= \boldsymbol{r}' \times \boldsymbol{r}' + \boldsymbol{r} \times \boldsymbol{r}'' \\ &= \boldsymbol{0} - \frac{k}{r^3} \boldsymbol{r} \times \boldsymbol{r} = \boldsymbol{0} \end{aligned}$$

3. $\boldsymbol{h}$ is constant. The relative angular momentum is $\mu \boldsymbol{h}$.

Orbital plane

- Since $h = \|\boldsymbol{h}\| > 0$ and $\boldsymbol{r} \cdot \boldsymbol{h} = 0$, the orbit lies in the plane perpendicular to $\boldsymbol{h}$. Assume $h > 0$ for $h = 0$ the motion is simply radial!
- Choose orthonormal axes in the plane oriented with $\theta' > 0$.

$$\boldsymbol{r} = r\boldsymbol{e}_r \qquad \boldsymbol{e}_r = (\cos\theta, \sin\theta) \qquad \boldsymbol{e}_\theta = (-\sin\theta, \cos\theta)$$

Polar velocity and acceleration

1. The unit vectors rotate with $\theta(t)$.

$$\boldsymbol{e}_r' = \theta' \boldsymbol{e}_\theta \qquad \boldsymbol{e}_\theta' = -\theta' \boldsymbol{e}_r$$

2. Differentiate $\boldsymbol{r} = r\boldsymbol{e}_r$ twice.

$$\begin{aligned} \boldsymbol{r}' &= r'\boldsymbol{e}_r + r\theta'\boldsymbol{e}_\theta \\ \boldsymbol{r}'' &= (r'' - r(\theta')^2)\boldsymbol{e}_r + (2r'\theta' + r\theta'')\boldsymbol{e}_\theta \end{aligned}$$

3. Compare with $\boldsymbol{r}'' = -k\boldsymbol{e}_r/r^2$.

Radial and angular equations

- Equate the two components.

$$r'' - r(\theta')^2 = -\frac{k}{r^2} \qquad r\theta'' + 2r'\theta' = 0$$

- Multiply the angular equation by r.

$$(r^2\theta')' = r^2\theta'' + 2rr'\theta' = 0$$

- Therefore

$$\boxed{r^2\theta' = h} \qquad \theta' = \frac{h}{r^2}$$

Equal areas in equal times

- A small angular displacement sweeps an area $dA = \frac{1}{2} r^2 \, d\theta$.

$$\frac{dA}{dt} = \frac{1}{2} r^2 \theta' = \frac{h}{2}$$

  Constant area rate is Kepler's second law [1].

- Constant angular momentum h gives motion once the orbit shape determined. The motion is faster at small separation

$$\theta' = \frac{h}{r^2}$$

Reciprocal separation

- Use angle as the independent variable and define

$$u(\theta) = \frac{1}{r} \qquad \theta' = hu^2$$

- Primes still denote time derivatives; write $du/d\theta$ explicitly.

$$\frac{d}{dt} = \theta' \frac{d}{d\theta} = hu^2 \frac{d}{d\theta}$$

- Since $r = u^{-1}$, the chain rule gives

$$r' = -u^{-2} \frac{du}{d\theta} \theta' = -h \frac{du}{d\theta}$$

Radial equation in angle

1. Differentiate $r' = -h\,du/d\theta$ with respect to time.

$$r'' = -h\frac{d^2u}{d\theta^2}\theta' = -h^2u^2\frac{d^2u}{d\theta^2}$$

2. Also $r(\theta')^2 = u^{-1}(hu^2)^2 = h^2u^3$.
3. Substitute into $r'' - r(\theta')^2 = -k/r^2$.

$$-h^2u^2\left(\frac{d^2u}{d\theta^2} + u\right) = -ku^2$$

4. Divide by $-h^2u^2$ to obtain

$$\boxed{\frac{d^2u}{d\theta^2} + u = \frac{k}{h^2}}$$

Solving the orbit equation

1. The solution is
$$u = \frac{k}{h^2} + C_1 \cos\theta + C_2 \sin\theta$$
2. Write the constants as an amplitude and a phase.
$$u = \frac{k}{h^2}\left[1 + \epsilon\cos(\theta - \theta_0)\right] \qquad \epsilon = \frac{h^2}{k}\sqrt{C_1^2 + C_2^2} \geq 0$$

$\epsilon > 0$ is the **Eccentricity**. θ_0 is the closest approach angle.

Polar orbit equation

- Choose $\theta = 0$ at closest approach and set $p = h^2/k$.

$$\boxed{r(\theta) = \frac{p}{1 + \epsilon \cos\theta}} \qquad p = \frac{h^2}{GM} > 0$$

- p is the **Semi-Latus Rectum**, the separation at $\theta = \pi/2$.
- For $0 \leq \epsilon < 1$, the denominator stays positive for every angle.
- This is an ellipse [1] for $\epsilon > 0$. It is a circle for $\epsilon = 0$.

Cartesian orbit equation

1. Let (x_1, x_2) be coordinates in the orbital plane.

$$x_1 = r\cos\theta \qquad x_2 = r\sin\theta \qquad r = \sqrt{x_1^2 + x_2^2}$$

2. Multiply the polar equation by its denominator.

$$r + \epsilon x_1 = p \qquad \sqrt{x_1^2 + x_2^2} = p - \epsilon x_1$$

3. Square and collect terms.

$$(1 - \epsilon^2)x_1^2 + x_2^2 + 2p\epsilon x_1 = p^2$$

Completing the square

1. For $0 \leq \epsilon < 1$, complete the square in x_1.

$$(1-\epsilon^2)\left(x_1 + \frac{p\epsilon}{1-\epsilon^2}\right)^2 + x_2^2 = \frac{p^2}{1-\epsilon^2}$$

2. Define the semi-major and semi-minor axes.

$$a = \frac{p}{1-\epsilon^2} \qquad b = \frac{p}{\sqrt{1-\epsilon^2}} = a\sqrt{1-\epsilon^2}$$

3. Divide to obtain the Cartesian ellipse equation.

$$\boxed{\frac{(x_1 + a\epsilon)^2}{a^2} + \frac{x_2^2}{b^2} = 1}$$

Focus and closest approach

- The ellipse center is $(-a\epsilon, 0)$, and the focal distance from the center is

$$\sqrt{a^2 - b^2} = a\epsilon$$

- The foci are $(0, 0)$ and $(-2a\epsilon, 0)$, so the force center lies at a focus.
- The nearest and farthest separations follow directly from the polar equation.

$$r_{\text{min}} = \frac{p}{1+\epsilon} = a(1-\epsilon) \qquad r_{\text{max}} = \frac{p}{1-\epsilon} = a(1+\epsilon)$$

- Therefore

$$\boxed{r(\theta) = \frac{a(1-\epsilon^2)}{1+\epsilon\cos\theta}}$$

Relative energy conservation

- Divide the relative energy by the reduced mass to define **Specific Orbital Energy**.

$$\mathcal{E} = \frac{E_{\text{rel}}}{\mu} = \frac{1}{2}\|\boldsymbol{r}'\|^2 - \frac{k}{r}$$

- Differentiate using $r' = \boldsymbol{r} \cdot \boldsymbol{r}'/r$.

$$\mathcal{E}' = \boldsymbol{r}' \cdot \left(-k\frac{\boldsymbol{r}}{r^3}\right) + \frac{kr'}{r^2} = 0$$

- In polar coordinates

$$\mathcal{E} = \frac{1}{2}(r')^2 + \frac{h^2}{2r^2} - \frac{k}{r}$$

Energy and eccentricity

1. From $u = (k/h^2)(1 + \epsilon\cos\theta)$ and $r' = -h\,du/d\theta$

$$r' = \frac{k\epsilon}{h}\sin\theta$$

2. Substitute r' and $u = 1/r$ into the energy.

$$\begin{aligned}\mathcal{E} &= \frac{k^2}{2h^2}\left[\epsilon^2\sin^2\theta + (1+\epsilon\cos\theta)^2 - 2(1+\epsilon\cos\theta)\right] \\ &= \frac{k^2}{2h^2}(\epsilon^2 - 1)\end{aligned}$$

3. Thus

$$\boxed{\epsilon^2 = 1 + \frac{2\mathcal{E}h^2}{k^2}}$$

Condition for an elliptical orbit

- For $h > 0$, negative energy gives $0 \leq \epsilon < 1$.

$$\mathcal{E} < 0 \quad \Longleftrightarrow \quad 0 \leq \epsilon < 1$$

- The circle is the case $\epsilon = 0$; $0 < \epsilon < 1$ gives a noncircular ellipse.
- Since $p = h^2/k = a(1 - \epsilon^2)$

$$\mathcal{E} = -\frac{k}{2a} \qquad E_{\text{rel}} = -\frac{Gm_1m_2}{2a}$$

- For $h > 0$, zero and positive energy give a parabola and hyperbola, respectively [1].

Ellipse from initial conditions

1. Given $\boldsymbol{r}(0)$ and $\boldsymbol{r}'(0)$, compute

$$h = \|\boldsymbol{r}(0) \times \boldsymbol{r}'(0)\| \qquad \mathcal{E} = \frac{1}{2}\|\boldsymbol{r}'(0)\|^2 - \frac{k}{\|\boldsymbol{r}(0)\|}$$

2. For $h > 0$ and $\mathcal{E} < 0$, find the ellipse parameters.

$$a = -\frac{k}{2\mathcal{E}} \qquad \epsilon = \sqrt{1 + \frac{2\mathcal{E}h^2}{k^2}} \qquad p = \frac{h^2}{k}$$

3. Initial position and velocity fix the orbital plane, travel direction, and phase θ_0.

Ellipses of both bodies

- In the center-of-mass rest frame, the reconstruction formulas give

$$\boldsymbol{r}_1 = -\frac{m_2}{M}\boldsymbol{r} \qquad \boldsymbol{r}_2 = \frac{m_1}{M}\boldsymbol{r}$$

- The bodies trace a scaled ellipse with eccentricity ϵ and focus at the center of mass.

$$a_1 = \frac{m_2}{M}a \qquad a_2 = \frac{m_1}{M}a \qquad a_1 + a_2 = a$$

Summary

- Central acceleration conserves angular momentum and fixes the orbital plane. [1].
- Reciprocal separation $1/r$ satisfies a linear equation in angle, giving

$$r(\theta) = \frac{a(1-\epsilon^2)}{1+\epsilon\cos\theta}$$

- Negative relative energy gives an ellipse (including $\epsilon = 0$ for a circle) [1].

References I

[1] Jaime Peraire and Sheila Widnall.
Lecture L15: Central force motion: Kepler's laws.
MIT OpenCourseWare, 16.07 Dynamics, 2008.
Version 1.2, Fall 2008; hosted in the Fall 2009 course; accessed 12 October 2026.